

Math 243: Homework 07

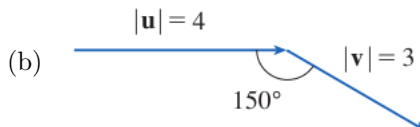
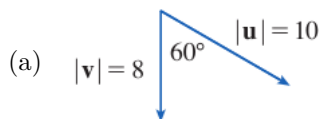
Due Monday, October 12

Problem 1. Compute the cross product $\mathbf{C} = \mathbf{A} \times \mathbf{B}$ and verify that $\mathbf{C}$ is orthogonal to both $\mathbf{A}$ and $\mathbf{B}$:

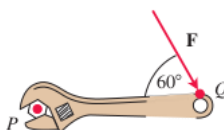
(a) $\mathbf{A} = \langle 2, 3, 0 \rangle$ and $\mathbf{B} = \langle 1, 0, 5 \rangle$

(b) $\mathbf{A} = \langle t, \cos(t), \sin(t) \rangle$ and $\mathbf{B} = \langle 1, -\sin(t), \cos(t) \rangle$

Problem 2. For both parts (a) and (b), find $|\mathbf{u} \times \mathbf{v}|$ and determine whether $\mathbf{u} \times \mathbf{v}$ is directed into the page or out of the page:



Problem 3. Consider a bolt at point P gripped by a wrench. A force $\mathbf{F}$ is applied at the endpoint Q of the wrench, as shown.



Let's assume that $|\overrightarrow{PQ}| = 0.25$ meters, and that the magnitude of the force is $|\mathbf{F}| = 120$ Newtons, which is about 30 lbs of force. Find the magnitude of the torque exerted by the force $\mathbf{F}$ on the bolt at point P .

Problem 4. For $\mathbf{A} = \langle 2, -1, 3 \rangle$ and $\mathbf{B} = \langle 2, 1, -1 \rangle$ compute both $\mathbf{A} \times \mathbf{B}$ and $\mathbf{B} \times \mathbf{A}$.

Problem 5. Find the area of the parallelogram with vertices $P = (1, 0, 2)$, $Q = (3, 3, 3)$, $R = (7, 5, 8)$ and $S = (5, 2, 7)$.

Problem 6. Compute the angles of the triangle formed by the points $A = (-1, 0)$, $B = (2, 1)$, and $C = (1, -2)$. *Hint: Use dot products.*

Problem 7. Find a nonzero vector orthogonal to the plane passing through the points $A = (1, 0, 1)$, $B = (-2, 1, 3)$, and $C = (4, 2, 5)$

Problem 8. A wrench 0.3 meters long lies along the positive y -axis and grips a bolt at the origin. Let $a > 0$ be a real number. A force $\mathbf{F} = \langle 0, \frac{3}{5}a, -\frac{4}{5}a \rangle$ is applied to the tip of the wrench handle.

(a) Sketch this problem set up.

(b) What is the magnitude of the force vector? Include units.

(c) Compute the torque vector $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$, using $\mathbf{r} = \langle 0, 0.3, 0 \rangle$ to represent the handle of the wrench. Torque here is measured in Newton-meters (Nm).

(d) Find the magnitude of the force needed to supply 100 Nm of torque to the bolt.

Problem 9. Find the unit tangent vector of the following curve

$$\mathbf{r}(t) = \left\langle t^2 - 2t, 1 + 3t, \frac{1}{3}t^3 + \frac{1}{2}t^2 \right\rangle$$

Problem 10. Find the parametric equation of the line through the points $(-8, 1, 4)$ and $(3, -2, 4)$.

Problem 11. Find an equation for the line segment joining the points $P = (-1, 2, -2)$ and $Q = (-3, 5, 1)$.

Problem 12. The planes given by the equations $5x + 8y - 2z = 2$ and $x + y - 2z = 2$ intersect to form a line. Find an equation of this line.